FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – VII
PAPER-1

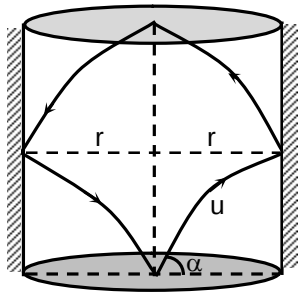
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**B, D**	19.	**A, D**	37.	**A, D**
2.	**B, C**	20.	**A, D**	38.	**B, C**
3.	**B, D**	21.	**B, C**	39.	**A, B**
4.	**A, D**	22.	**A, B, D**	40.	**A, B, C**
5.	**B, C**	23.	**A, B, D**	41.	**A, B, D**
6.	**A, C, D**	24.	**A, B**	42.	**A, D**
7.	**B, C**	25.	**A, B, C**	43.	**A, B, D**
8.	**B**	26.	**A**	44.	**A**
9.	**A**	27.	**D**	45.	**B**
10.	**D**	28.	**A**	46.	**D**
11.	**D**	29.	**C**	47.	**B**
12.	**C**	30.	**D**	48.	**A**
13.	**B**	31.	**B**	49.	**C**
14.	**4**	32.	**5**	50.	**6**
15.	**3**	33.	**4**	51.	**2**
16.	**2**	34.	**4**	52.	**7**
17.	**2**	35.	**4**	53.	**3**
18.	**2**	36.	**3**	54.	**9**

Physics

PART – I

SECTION – A

1. B, D



During collision u_x will not change and range will be 4r.

$\therefore \quad 4r = (u\cos\alpha)T$

$\Rightarrow \quad u = \dfrac{4r}{T\cos\alpha}$

As particle will not be able to reach highest position of standard projectile path.

$\dfrac{2u\sin\alpha}{g} > T \quad \Rightarrow \dfrac{2u\sin\alpha}{g} > \dfrac{4r}{u\cos\alpha}$ From (1)

$\Rightarrow \quad u > 2\sqrt{\dfrac{gr}{\sin 2\alpha}}$

2. B, C

$T = 2\pi\sqrt{\dfrac{\ell}{g}} \Rightarrow g \propto \ell T^2$

$\Rightarrow \quad \dfrac{\Delta g}{g} = \left(\dfrac{\Delta\ell}{\ell} + \dfrac{2\Delta T}{T}\right)$

Therefore $P = \left(\dfrac{\Delta\ell}{\ell} + \dfrac{2\Delta T}{T}\right)100\%$

$\Rightarrow \quad P_1 = \left(\dfrac{0.1}{100} + \dfrac{2(1)}{400}\right)\times 100 = 0.6\%$

$P_2 = \left(\dfrac{0.1}{400} + \dfrac{2(1)}{400}\right)\times 100 = 0.42\%$

$P_3 = \left(\dfrac{0.1}{100} + \dfrac{2(1)}{200}\right)\times 100 = 1.1\%$

$P_4 = \left(\dfrac{0.1}{400} + \dfrac{2(1)}{800}\right)\times 100 = 0.28\%$

3. B, D

$B = \dfrac{\mu_0 I}{\ell} = \dfrac{\mu_0}{\ell}\left[(\sigma)\dfrac{(2\pi R)\ell\omega}{2\pi}\right]$

$= \mu_0\sigma R\omega$

$= \mu_0\sigma Rkt \Rightarrow B \ \alpha \ 't'$

Consider loop of radius 'r' $r < R$

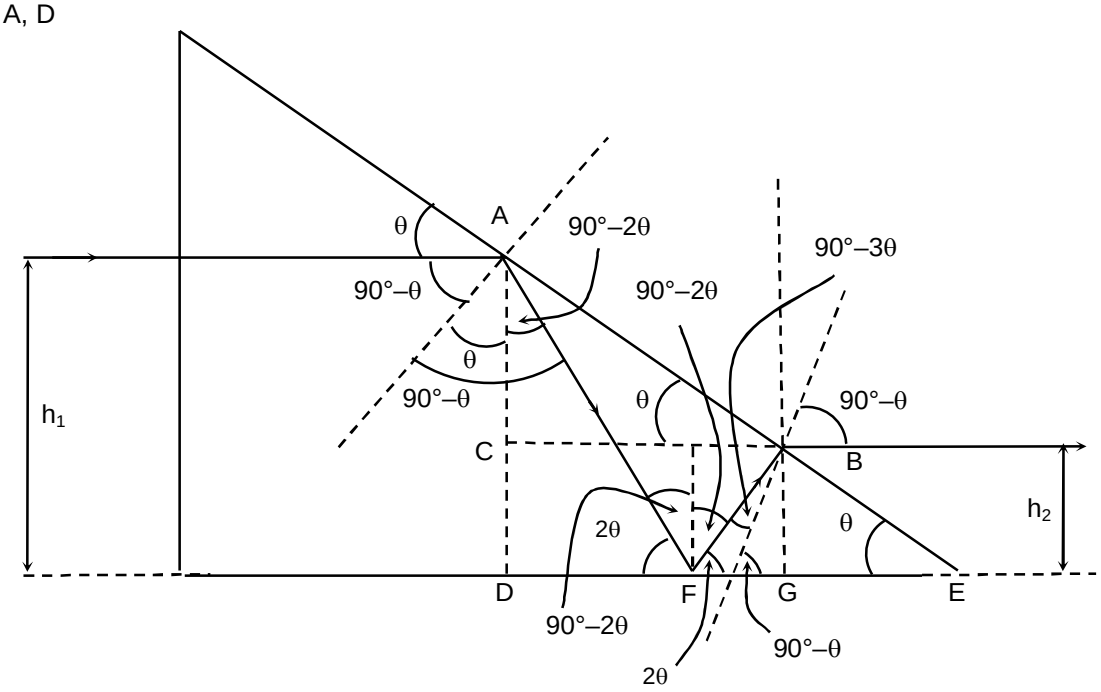
$E(2\pi r) = \pi r^2 \dfrac{dB}{dt}$

$E = \dfrac{r}{2}(\mu_0\sigma Rk) \qquad E \propto r$

For,

Energy density $= \dfrac{B^2}{2\mu_0}$

4. A, D



$AC = h_1 - h_2$

$$\tan\theta = \frac{AC}{CB}$$

$$= \frac{h_1 - h_2}{DF + FG}$$

$$\tan\theta = \frac{h_1 - h_2}{h_1\cot 2\theta + h_2\cot 2\theta}$$

$$\Rightarrow \quad \frac{\tan\theta}{\tan 2\theta} = \frac{h_1 - h_2}{h_1 + h_2}$$

$$\Rightarrow \quad \frac{h_1}{h_2} = \frac{\sin\theta}{\sin 3\theta}$$

$$\Rightarrow \quad \frac{h_2}{h_1} = \frac{\sin 3\theta}{\sin\theta} = 4\cos^2\theta - 1 \qquad (1)$$

Using Snel's law at 'B'

$$\mu\sin(90° - 3\theta) = \sin(90° - \theta)$$

$$\Rightarrow \quad \frac{1}{\mu} = \frac{\cos 3\theta}{\cos\theta} = 4\cos^2\theta - 3 \qquad (2)$$

From (1) and (2) $\frac{h_1}{h_2} = \frac{\mu}{1+2\mu}$

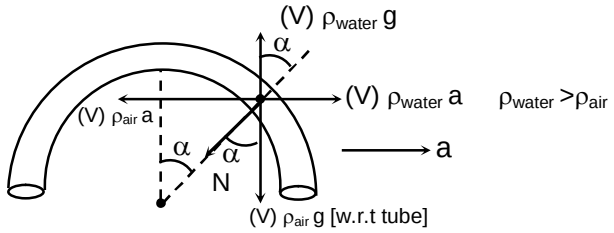
5. B, C

Use condition of equilibrium

$N\cos\alpha = v(\rho_{water} - \rho_{air})g$

$N\sin\alpha = V(\rho_{water} - \rho_{air})a$

$$\Rightarrow \quad \tan\alpha = \frac{a}{g}$$



6. A, C, D

$U = 3PV$

$nC_vT = 3PV$

$\Rightarrow \quad \gamma = \frac{4}{3}$

Tri-atmoic gas

$C_p = \frac{\gamma R}{\gamma - 1}$

$\Rightarrow \quad C_p = 4R$

7. B, C

Block will move between $\frac{-\alpha}{2} \& \frac{\alpha}{2}$.

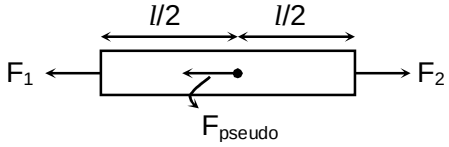
$T = 4\left(\frac{T_0}{6}\right) = \frac{2}{3}T_0$

If collision with atleast one wall is inelastic then amplitude will reduce and time period will remain same as T_0.

8. B
9. A
10. D
11. D
12. C
13. B

SECTION – C

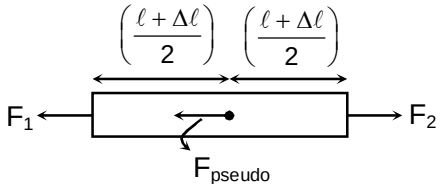
14. 4



From this frame, $W_{pseduo} = 0$

$W_1 = F_1 \times \left(\frac{\Delta \ell}{2}\right)$

And $W_2 = F_2 \times \left(\frac{\Delta \ell}{2}\right)$



Thus, w.e.f. from COM frame

$W_1 + W_2 + W_{pseudo} = U + \kappa + \Delta$ [$\therefore$ k will remains zero finally from COM frame]

$\Rightarrow \quad W_1 + W_2 = U + \Delta$

$\Rightarrow \quad F_1.\frac{\Delta \ell}{2} + F_2.\frac{\Delta \ell}{2} = U + \Delta$

$\Rightarrow \quad (F_1 + F_2)\frac{\Delta \ell}{2} = U + \Delta$

So, Let's calculates heat now

$$\frac{1}{2}\left[(F_1+F_2)\frac{(F_1+F_2)\ell}{2AY}\right]=\frac{(F_1^2+F_2^2+F_1F_2)\ell}{6AY}+\Delta$$

$$\Rightarrow \quad \Delta=\frac{(F_1^2+F_2^2+4F_1F_2)\ell}{12AY}$$

Putting $F_1 = F$; $F_2 = 2F$, we get $\Delta=\frac{13F^2\ell}{12AY}$

$\Rightarrow$ a = 1; b = 3

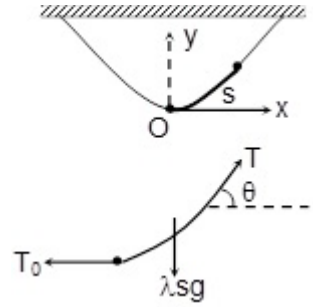
$\Rightarrow$ a + b = 4

15. 3

$T\sin\theta=\lambda sg$ . ...(1)

$T\cos\theta=T_0$ ....(2)

(i) and (ii) $\Rightarrow \tan\theta=\lambda sg/T_0 \quad \Rightarrow \quad S=\left(\frac{T_0}{\lambda g}\right)\tan\theta.$



R.O.C. $=\frac{ds}{d\theta}$

$$\Rightarrow \quad R=\left(\frac{T_0}{\lambda g}\right)\sec^2\theta$$

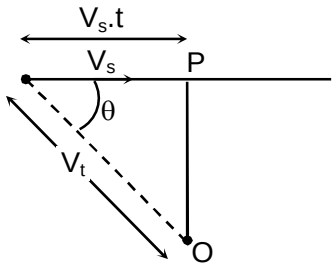
Putting $T_0 = 90$; $\lambda = 4; g = 10$ and $\theta = 30°$

R = 3 m

16. 2

The observer will hear sound, which has been emitted when source is slightly before point P.

Distance between two consecutive pubes emitted towards observer is $\frac{V-V_s\cos\theta}{f}$, as shown below.



$$\Rightarrow \lambda'=\frac{V-V_s\cos\theta}{f}$$

Putting $\cos\theta=\frac{V_s}{V}$

We get $\lambda'=\frac{V^2-V_s^2}{Vf}$

Putting the given values we get

$\lambda'=\frac{2}{11}$m

Hence n = 2

17. 2

$$mv+(m+\Delta m)g\Delta t=(m+\Delta m)(v+\Delta v)$$

$$\Rightarrow \quad mv+mg\Delta t+\Delta mg\Delta t=mv+m\Delta v+\Delta m.v+\Delta m.\Delta v$$

$$\Rightarrow \quad mg\Delta t+\Delta mg\Delta t=m\Delta v+\Delta m.v+\Delta m.\Delta v \qquad ....(1)$$

Also, $\frac{\Delta m}{\Delta t}=kmv$ (given)

$$\Rightarrow \quad \Delta m=kmv\Delta t$$ Putting in (1)

We get $mg\Delta t+kmvg\Delta t^2=m\Delta v+kmv^2\Delta t+kmv\Delta t.\Delta v$

$$\Rightarrow \quad mg\Delta t=m\Delta v+kmv^2\Delta t$$ (Neglecting second differentials)

$\Rightarrow \quad \Delta v = (g - kv^2)\Delta t \quad \Rightarrow \quad \frac{\Delta v}{\Delta t} = g - kv^2$

$\Rightarrow \quad \frac{dv}{dt} = g - kv^2$

At terminal velocity $g - kv^2 = 0$

$\Rightarrow \quad v = \sqrt{\frac{g}{k}}$

Putting g = 10; k = 2.5 $\Rightarrow$ v = 2 m/s

18. 2

Let $y = A\sin(\omega t - kx)$

P.E. stored in small element dx is

$dU = \frac{T}{2}\left(\frac{\partial y}{\partial x}\right)^2 .dx = \frac{\mu v^2}{2}\left(\frac{\partial y}{\partial x}\right)^2 dx$

$= \frac{\mu}{2}\left[v.\frac{\partial y}{\partial x}\right]^2 dx$

Since, $\frac{\partial y}{\partial t} = -v\frac{\partial y}{\partial x}$

Hence, $dU = \frac{\mu}{2}.\left(\frac{\partial y}{\partial t}\right)^2 dx$

Also, KE stored in small element 'dx' is

$dK = \frac{1}{2}(\mu dx)\left(\frac{\partial y}{\partial t}\right)^2$

$\Rightarrow$ Total energy dE = dU + dK = $\mu\left(\frac{\partial y}{\partial t}\right)^2 dx$

$\frac{\partial y}{\partial t} = A\omega\cos(\omega t - kx)$

$\Rightarrow \quad dE = \mu\omega^2 A^2\cos^2(\omega t - kx)dx$

$\Rightarrow \quad E = \mu\omega^2 A^2 \int_{x=0}^{\lambda} \cos^2(\omega t - kx)dx = \frac{\mu\omega^2 A^2\lambda}{2}$

Hence b = 2

Chemistry

PART – II

SECTION – A

19. AD
Addition of $CaCO_3$ does not affect equilibrium as it is solid and inert gas addition at constant volume also does not affect the equilibrium but as reaction is endothermic and gaseous Stoichiometry of product is more than reactants that's why A and D are correct

20. AD
Both A and D are correct as both follows Hund's rule and B, C do not

21. BC
H-atom in NaH_2PO_2 is bonded with P (less electronegative), cannot involve in H-bonding and so also $(CH_3)_3BO_3$

22. ABD
At pH < isoelectric pH, $-NH_2$ group is protonated to $-\overset{+}{N}H_3$ and the sol is positively charged. At pH > isoelectric pH, –COOH is deprotonated to $-COO^-$ and the sol is negatively charged. At isoelectric pH both groups are equally ionised, the sol particles carry no net charge.

23. ABD
Sodium nitrate on decomposition upto 500ºC gives $NaNO_2$ and oxygen
$2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2 \uparrow$
While at higher temperature (i.e. above to 800ºC), $NaNO_2$ further decomposes into Na_2O, N_2 and O_2.
$2NaNO_2 \xrightarrow{800°C} Na_2O + 3/2O_2 \uparrow + N_2 \uparrow$

24. AB
$[K_2HgI_4 + KOH]$ and $[HgCl_2 + KI(\text{excess}) + KOH]$ are Nessler's reagent which gives brown precipitate of $H_2N-Hg-O-Hg-I$ (iodide of Million's base).

25. ABC
Since there are no ions involved in this compound, the Cl^- atoms are covalently bonded to Pd in the coordination sphere
(d) No precipitate of AgCl would be formed as chlorine atoms are non-ionisable.

26. A
Tollen's test is given by all type of aldehyde by action of ammonical silver nitrate and observation is formation of silver mirror

27. D
Red ppt is observed by action of Fehling solution on aliphatic aldehyde

28. A
Tollen's test, Fehling's test and Haloform test all three involve oxidation of organic compound but action of Fehling solution on terminal alkynes do not

29. C

30. D

31. B

Sol. (29 to 31)

Crystal system	Axial length & angle	Type of unit cell
Cube	$a = b = c$ $\alpha = \beta = \gamma$	Primitive, face centred & body centered
Orthorhombic	$a \neq b \neq c$ $\alpha = \beta = \gamma$	Primitive, Face centred, body centered & end centered
Triclinic	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma$	Only Primitive
Monoclinic	$a \neq b \neq c$ $\alpha = \beta \neq \gamma$	Primitive, End centred

SECTION – C

32. 5

Moles ($KMnO_4$) $\times$ 5 = mole (SnC_2O_4) $\times$ 4 (n factor of SnC_2O_4)

$SnC_2O_4 \xrightarrow{KMnO_4} Sn^{4+} + CO_2$

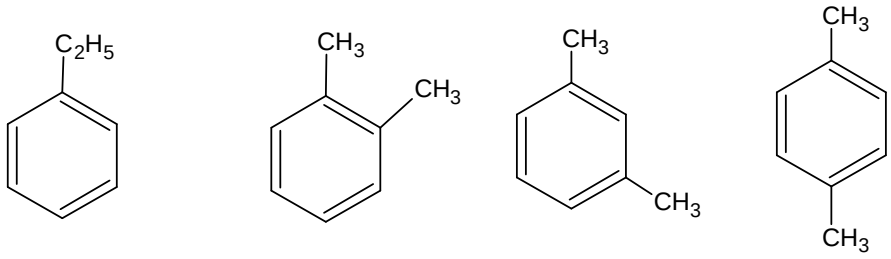
33. 4

One triangular face will be found in place of each corner of tetrahedron

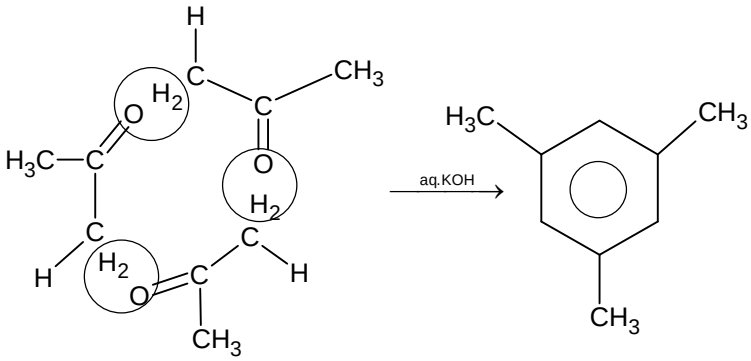
34. 4

$^{235}_{92}U + ^{1}_{0}n \rightarrow {}^{140}_{54}Xe + {}^{92}_{38}Sr + 4^{1}_{0}n$

35. 4



36. 3



Mathematics

PART – III

SECTION – A

37. A, D

Sol. $y^2 = 16x \Rightarrow a = 4$

Slope of tangents at t is $\frac{1}{t}$

$\tan\theta_1 = \frac{1}{t_1}$, $\tan\theta_2 = \frac{1}{t_2}$; $\tan\theta_3 = \frac{1}{t_3}$

t_1, t_2, t_3 are in A. P. with $d = 3$

$t_2 - t_1 = 3$

$t_3 - t_2 = 3$

Area of $\Delta PQR = \frac{1}{2}\begin{vmatrix} at_1t_2 & a(t_1+t_2) & 1 \\ at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \end{vmatrix}$

$= a^2d^3 = 16 \times 27 = 432$

Area of ΔABC is $= 2.\Delta PQR$

= 864.

38. B, C

Sol. For. A, $|z+1| \le 2 + Re(z)$

$\Rightarrow y^2 \le 2\left(x + \frac{3}{2}\right)$

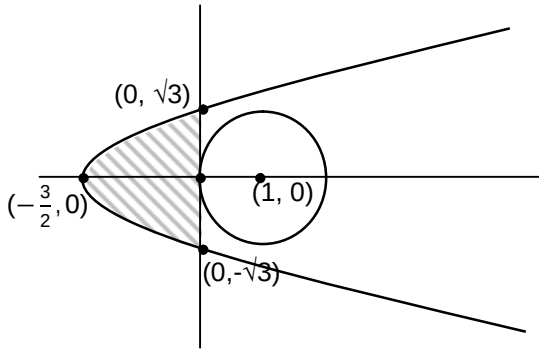
For B, $|z-1| \ge 1 \qquad \Rightarrow (x-1)^2 + y^2 \ge 1$

For C

$|z-1|^2 \ge |z+1|^2$

$\Rightarrow (z-1)(\bar{z}-1) \ge (z+1)(\bar{z}+1)$

$\Rightarrow z + \bar{z} \le 0 \qquad \Rightarrow x \le 0$



B. Required area $2\int_{-\frac{3}{2}}^{0}\sqrt{2(x+\frac{3}{2})}\,dx = 2\sqrt{3}$

C. $z = -\frac{3}{2} + i0$ is complex number in region $A \cap B \cap C$ having max amplitude

D. $(-1,0),(-1,1),(-1,-1),(0,0),(0,1)(0,-1)$, are point but $z = -1$ is not in domain of C

so total points 5

39. A, B

Sol. $AB = \begin{bmatrix} 3ax^2 & 3bx^2 & 3cx^2 \\ a & b & c \\ 6ax & 6bx & 6cx \end{bmatrix}$

Now $t_r(AB) = t_r(c)$

$\Rightarrow 3ax^2 + b + 6cx = (x+2)^2 + 2x + 5x^2 : \forall x \in R$

$a = 2, c = 1, b = 4$

$I = \int_0^\infty \frac{\ln x}{x^2+2x+4}dx$

Put $x = 2t$

$dx = 2dt$

$= \int_0^\infty \frac{\ln(2t)}{4t^2+4t+4}.2dt$

$= \int_0^\infty \frac{2.\ln 2}{4(t^2+t+1)}dt + \int_0^\infty \frac{2\ln t}{4(t^2+t+1)}dt$

Put $t = \frac{1}{y}$ in second Integral we get zero

$= \frac{1}{2}\int_0^\infty \frac{\ln 2}{t^2+t+1}dt + 0$

$= \frac{\pi \ln 2}{3\sqrt{3}}$

$p = 2, q = 27$

$p + q = 29$

40. A, B, C

Sol. $\text{Probability} = \frac{3(n-4)}{4(n-1)}$: If n is even

$\text{Probability} = \frac{3(n-3)}{4(n-2)}$: If n is odd

41. A, B, D

Sol. $g(x) = 4$

$x^3\left(f'(t)-2\right) + x^2 f''(t) + 4x\left(f(0)+6\right) + 4 = 4$ for $\forall x$

$\Rightarrow x^3(f'(t)-2) + x^2 f''(t) + 4x(f(0)+6) = 0$ for $\forall x$

Hence $f'(t) - 2 = 0$ and $f''(t) = 0$

and $f(0) + 6 = 0 \Rightarrow f(0) = -6$

So, $f(t) = 2t - 6$

$$h(x) = \begin{cases} -x^2 + 8x & ;0 \le x < 4 \\ x^2 - 8x + 32 & ;4 \le x \le 6 \\ (x-6)^2 + 20 & ;6 < x \le 12 \end{cases}$$

42. A, D

Sol. $(A)\left(f^{-1}g^{-1}(x)\right) = \left(g(f(x))\right)^{-1}$

$\frac{d}{dx}\left(f^{-1}g^{-1}(x)\right)\Big|_{\text{at } x=-12} = \frac{1}{\frac{d}{dx}\left(gof(x)\right)_{\text{at } x=-1}}$

(D) Area will be same as bounded by $y = f(x), y = g(x)$

$\Rightarrow \int_{-1}^{1}\left|x^3 - x\right|dx = \frac{1}{2}$

43. A, B, D

Sol. $\tan\theta = \frac{2y}{2x} = \frac{y}{x}$

$\tan\alpha = \frac{y}{2x}$

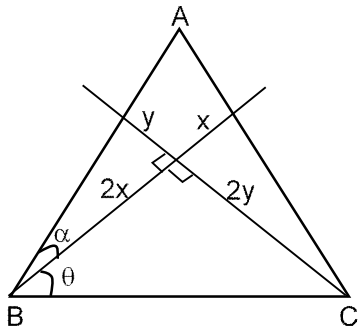
$\tan B = \frac{\frac{y}{x} + \frac{y}{2x}}{1 - \frac{y^2}{2x^2}} = \frac{3xy}{2x^2 - y^2}$

$\cot B = \frac{2x^2 - y^2}{3xy}$;

Similarly,

$\cot C = \frac{2y^2 - x^2}{3xy}$

$\cot B + \cot C = \frac{x^2 + y^2}{3xy} \geq \frac{2}{3}$

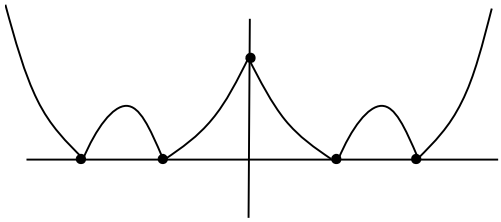


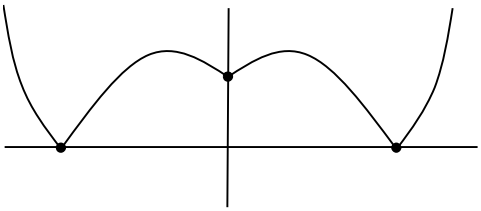
44. A

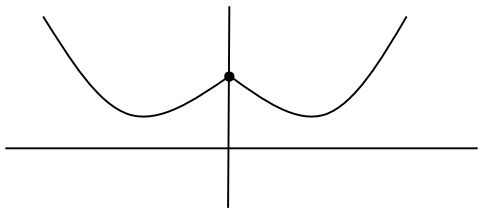
45. B

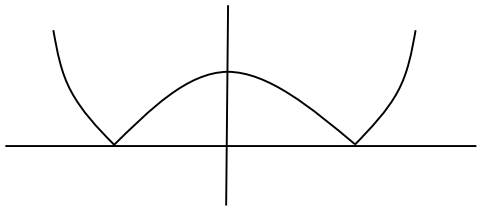
46. D

Sol.

I.  4 roots, 5 points not derivative

II.  2 roots, 3 points not derivative

III.  No root's, 1 point of non-derivative

IV.  2 roots, 2 points non-derivative

47. B
48. A
49. C

Sol. Point $(1, a_1, 1)$ lies on s_1 and s_2

$2 + 3a_1 + 4 = 9 \Rightarrow a_1 = 1$

$(b_1, b_2, 1)$ is $\perp$ to normal of planes s_1, s_2 and s_3

$2b_1 + 3b_2 + 4 = 0$

$4b_1 + 5b_2 + 6 = 0$

$\Rightarrow b_1 = 1, b_2 = -2, a = 2, b = 3$

SECTION – C

50. 6

Sol. $\frac{S_n}{\Delta_n} = \frac{2n+5}{3n+2} = \lambda$

$S_n = \lambda n(2n+5) : \Delta_n = \lambda n(3n+2)$

$T_n = S_n - S_{n-1} : t_n = \Delta_n - \Delta_{n-1}$

$T_n = \lambda(4n+3) : t_n = \lambda(6n-1)$

$T_{11} = \lambda(47)$

$t_{10} = \lambda(59)$

$\frac{T_{11}}{t_{10}} = \frac{47}{59} = \frac{m_1}{m_2}$

$\frac{m_2 - m_1}{2} = \frac{59 - 47}{2} = 6$

51. 2

Sol. $\text{cosec}^{-1}(\text{cosec}2) > x^2 - 3x$

$\pi - 2 > x^2 - 3x$ give integral solution of 1,2, 3.

$4020 = 2^2.3.5.67$ given 24 divisors'

Now, $f(i) \le f(j) \forall i < j$

${}^{24+2}C_3 = 2600$

52. 7

Sol. $P\left(\alpha, \frac{1}{1+\alpha^2}\right)$ lie on

$y = \frac{1}{1+x^2}$

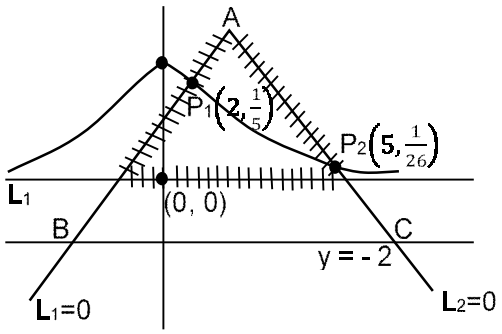
On solving $y = \frac{1}{1+x^2}$ with L_1

$P_1\left(2, \frac{1}{5}\right)$ ...(1)

And with L_2 , $P_2\left(5, \frac{1}{26}\right)$ ...(2)

From (1) and (2)

$2 \le \alpha \le 5$



53. 3

Sol. Let k be number of page that was counted twice then $0 < K < n+1$

$0+1+2+3+.....n < 1986 < 1+2+....(n+1)$

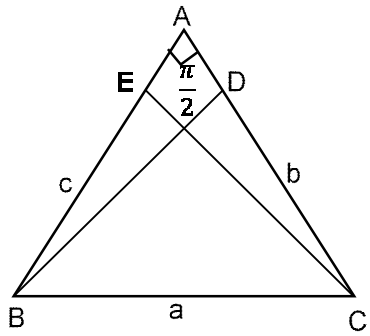
$n(n+1) < 3972 < (n+1)(n+2)$

On solving $n = 62$

So, $k = 1986 - \frac{62\times 63}{2} = 1986 - 1953 = 33$

54. 9

Sol. We have $\frac{a}{R} = 2 : \frac{b}{R} = \frac{3}{2}$



$\frac{2R\sin A}{R} = 2$

$\sin A = 1 \Rightarrow \angle A = \frac{\pi}{2}$

And $\sin B = \frac{3}{4}$

$c^2 = 4R^2 - \frac{9R^2}{4}$

$\Rightarrow c = \frac{\sqrt{7}}{2}R$

$l_1 = \frac{2ac}{a+c}\cos\frac{B}{2}$

$l_2 = \frac{2ab}{a+b}\cos\frac{C}{2}$

$\frac{l_1}{l_2} = \left(\frac{a+b}{a+c}\right)\frac{c}{b}.\sqrt{\frac{1+\cos B}{1+\cos C}}$

$= \frac{c(a+b)}{b(a+c)}.\sqrt{\frac{1+\frac{c}{a}}{1+\frac{b}{a}}}$

$= \frac{c}{b}.\sqrt{\frac{a+b}{a+c}}$

Now put,

$a = 2R,$

$b = \frac{3}{2}R,$

$c = \frac{\sqrt{7}}{2}R$

$\frac{l_1}{l_2} = \frac{7}{3}\left(\sqrt{\frac{4-\sqrt{7}}{16-7}}\right) = \frac{7}{9}.\sqrt{\frac{8-2\sqrt{7}}{2}} = \frac{7}{9.\sqrt{2}}(\sqrt{7}-1)$